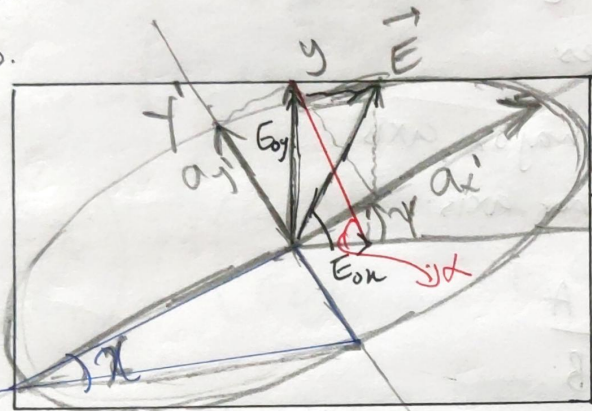


When $E_1 = E_2$ with $\delta = 90^\circ$, then the wave is circularly polarized.

But in many cases, axis of ellipse will not align with axis of detector. $\therefore$ to understand inherent nature of polarization, we need to know the intrinsic parameters of the ellipse too.



$$\tan \chi = \pm \frac{b}{a}$$

$$\tan \alpha = \frac{E_{0y}}{E_{0x}}$$

$$0 \leq \alpha \leq \pi/2$$

$$\vec{E} = \hat{x} E_{0x} \cos(\omega t - kz + \delta_1) + \hat{y} E_{0y} \cos(\omega t - kz + \delta_2)$$

Now this has to be written in $x'y'$ coordinates.

A vector on x,y axis will be rotated to x',y' as follows:

$$E_{x'} = E_{0x} \cos(\omega t - kz + \delta_1)$$

$$E_{y'} = E_{0y} \cos(\omega t - kz + \delta_2)$$

$$\begin{bmatrix} E_{x'} \\ E_{y'} \end{bmatrix} = \begin{bmatrix} \cos \psi & \sin \psi \\ -\sin \psi & \cos \psi \end{bmatrix} \begin{bmatrix} E_x \\ E_y \end{bmatrix}$$

Then E_x, E_y can be written in terms of E_x' & E_y' as

$$\begin{bmatrix} E_x \\ E_y \end{bmatrix} = \begin{bmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{bmatrix} \begin{bmatrix} E_x' \\ E_y' \end{bmatrix}$$

$$\text{Then } E_x = E_x' \cos \psi - E_y' \sin \psi$$

$$E_y = E_x' \sin \psi + E_y' \cos \psi$$

Substituting this in the ellipse equation, & grouping my the $E_x' E_y'$ terms & setting the equal to 0 [the curve ^{for} the ellipse to align with the measurement reference, the cross term must be 0].

$$\Rightarrow \frac{E_x' E_y' \sin 2\psi}{E_{ox}^2} + \frac{E_x' E_y' \sin 2\psi}{E_{oy}^2} - \frac{2 E_x' E_y' \cos 2\psi \cos \delta}{E_{ox} E_{oy}} = 0$$

Simplifying this algebraically, gives

$$\left[\frac{E_{ox}^2 - E_{oy}^2}{E_{ox}^2 \cdot E_{oy}^2} \right] \cdot \sin 2\psi = \frac{2 \cos 2\psi \cdot \cos \delta}{E_{ox} \cdot E_{oy}}$$

$$\Rightarrow \tan 2\psi = \left[\frac{2 E_{ox} \cdot E_{oy}}{E_{ox}^2 - E_{oy}^2} \right] \cos \delta$$

$$\text{but } \tan \alpha = \frac{E_{oy}}{E_{ox}}$$

$$\Rightarrow \boxed{\tan 2\psi = \tan 2\alpha \cdot \cos \delta}$$

For deriving χ , we start with

$$\tan \chi = \frac{\pm b}{a}.$$

~~But~~, To proceed further we invoke the rotational invariants of the ellipse. There are

$$\textcircled{1} \quad E_{0x}^2 + E_{0y}^2 = a^2 + b^2 = S_0$$

~~② To write the ellipse in its~~

② Area of the ellipse must remain the same. From this we get,

$$ab = E_{0x} \cdot E_{0y} \cdot \sin \delta$$

Taking the ratio of ② & ①, we will get

$$\frac{\textcircled{2}}{\textcircled{1}} = \frac{ab}{a^2 + b^2} \cdot \text{Why the fact } \tan \chi = \frac{b}{a},$$

$$\text{we can write } \frac{ab}{a^2 + b^2} = \frac{a}{\sqrt{a^2 + b^2}} \cdot \frac{b}{\sqrt{a^2 + b^2}}$$

$$\frac{ab}{a^2 + b^2} = \sin \chi \cdot \cos \chi$$

Now, Multiplying each side by 2, we get

$$\sin 2\chi = \frac{2ab}{a^2 + b^2}$$

However, $ab = E_{0x} E_{0y} \sin \delta$

$$\Rightarrow \sin 2\alpha = \frac{2 E_{0x} E_{0y} \sin \delta}{E_{0x}^2 + E_{0y}^2}$$

$$= \frac{2 (E_{0y}/E_{0x}) \sin \delta}{1 + (E_{0y}/E_{0x})^2}$$

$$\sin 2\alpha = \frac{2 \tan \alpha}{1 + \tan^2 \alpha} \cdot \sin \delta$$

$$\sin 2\alpha = \tan 2\alpha \cdot \sin \delta$$

↑
these invert the phase geometry & aux.
angle of the meas. plane to the ellipticity.

Now, the total intensity of light, is

$$S_0 = E_x^2 + E_y^2 \rightarrow \text{this is also called}$$

as 'I' in Stokes parameters

we can write 3 more parameters this way. They are:

$$S_1 = E_x^2 - E_y^2 \quad \begin{array}{l} \text{+ve when close to} \\ \text{horizontal} \\ \text{-ve when close to} \\ \text{vertical} \end{array}$$

$$S_2 = 2 E_x \cdot E_y \cdot \cos \delta \quad - 45^\circ / 135^\circ$$

$$S_3 = 2 E_x E_y \sin \delta \quad - \text{RCP/LCP}$$

If we normalize the ~~Stokes~~ Stokes parameters w.r.t. S_0 , then

$$S_1 = \frac{E_{0x}^2 - E_{0y}^2}{E_{0x}^2 + E_{0y}^2} = \frac{1 - \tan^2 \alpha}{1 + \tan^2 \alpha} = \cos 2\alpha$$

$$S_2 = \frac{2 E_{0x} E_{0y} \cos \delta}{E_{0x}^2 + E_{0y}^2} = \sin 2\alpha \cos \delta$$

$$S_3 = \frac{2 E_{0x} E_{0y} \sin \delta}{E_{0x}^2 + E_{0y}^2} = \sin 2\alpha \sin \delta$$

But this has to be in terms of ψ (polarization angle) and $\chi \rightarrow$ ellipticity angle

From prev. trig relations,

$$S_3 = \sin 2\chi$$

$$S_2 = \sin 2\psi \cos 2\chi$$

$$S_1 = \cos 2\psi \cos 2\chi$$

$\rightarrow$ These are like spherical polar coordinates

Summarizing

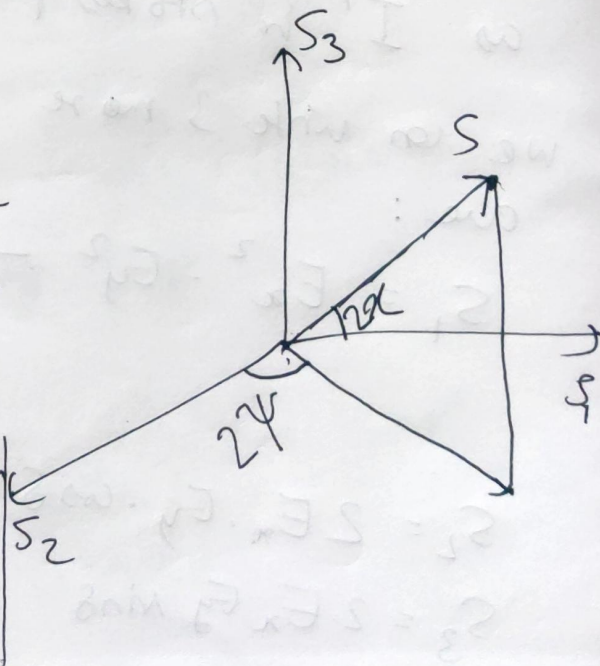
$$S_0 = 1$$

$$S_1 = \cos 2\psi \cos 2\chi$$

$$S_2 = \sin 2\psi \cos 2\chi$$

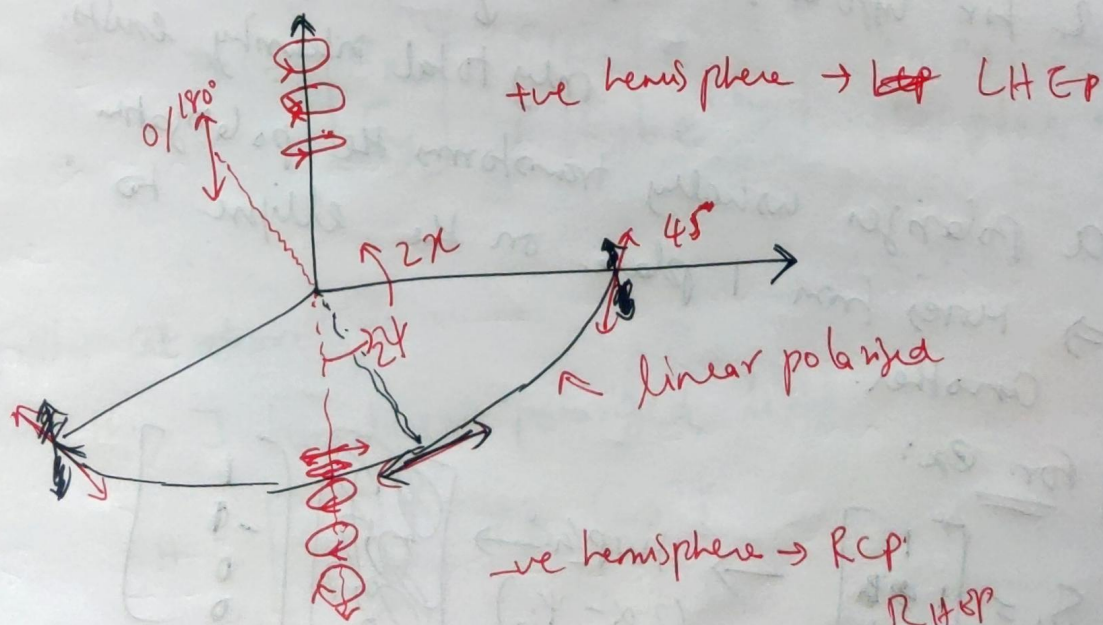
$$S_3 = \sin 2\chi$$

All possible states of polarization will form a sphere. Called the "Poincare Sphere"



$$\psi \rightarrow -\frac{\pi}{2} \text{ to } \frac{\pi}{2}, \quad \chi = -\frac{\pi}{4} \text{ to } \frac{\pi}{4}$$

	ψ	χ	Polarization
	$-\pi/2$	$-\pi/4$	$S_1=0, S_2=0, S_3=-1 \rightarrow \text{RCP}$
	$\pi/2$	$\pi/4$	$S_1=0, S_2=0, S_3=1 \rightarrow \text{LCP}$
	$-\pi/2$	0	$S_1=-1, S_2=0, S_3=0$
	$\pi/2$	0	$S_1=1, S_2=0, S_3=0$
	0	0	$S_1=1, S_2=0, S_3=0$
	$+\pi/4$	0	$S_1=0, S_2=+1, S_3=0$
	$-\pi/4$	0	$S_1=0, S_2=-1, S_3=0$



Sometimes there may exist other polarizations ~~in the~~ contributing to the total light

$$S_1^2 + S_2^2 + S_3^2 < S_0^2$$

~~tan 2ψ~~

$$\tan 2\psi = \frac{S_2}{S_1}$$

$$\psi = \frac{1}{2} \tan^{-1} \frac{S_2}{S_1} \rightarrow \text{polarization angle}$$

$$\chi = \frac{1}{2} \sin^{-1} (S_3)$$

In radio astronomy we will almost always encounter only partial polarization, with rare cases of full polarization.

$$S_{\text{partial}} = p S_{\text{pol}} + (1-p) S_{\text{unpol}}$$

$$p \rightarrow \frac{\sqrt{S_1^2 + S_2^2 + S_3^2}}{S_0}, \quad 0 < p < 1$$

for unpolarized light, $S = \begin{bmatrix} S_0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ $\downarrow$ only total intensity exists.

A polarizer usually transforms the polarization $\Rightarrow$ Moves from 1 place on the ellipse to another.

eg For ex:

$$S_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\xrightarrow{\text{HWP}(45^\circ)} \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

$(2\alpha - \psi_{in})$
 $2(\psi - \alpha)$
 $90^\circ \rightarrow$ goes around the equator.

Lecture 3

We will start with the ~~the Stokes parameters~~ electric field version of the Stokes parameters.

$$S_0 = E_x^2 + E_y^2$$

$$S_1 = E_x^2 - E_y^2$$

$$S_2 = 2 E_x E_y \cos \delta$$

$$S_3 = 2 E_x E_y \sin \delta$$

Let us suppose we have an initial state of the electric fields, E_x & E_y . Let's normalize this so that

$$E_x = 1, E_y = 0 \rightarrow \text{HP}$$

$$E_y = 1, E_x = 0 \rightarrow \text{VP. and so on. If we write}$$

the vector

$$\vec{E} = \begin{Bmatrix} E_x \\ E_y \end{Bmatrix} e^{i(\omega t - kz)}$$

$$J = \begin{bmatrix} E_x \\ E_y \end{bmatrix} = \begin{bmatrix} |E_x| e^{i\delta_x} \\ |E_y| e^{i\delta_y} \end{bmatrix}$$

δ_x, δ_y are the phase diff between the two components.

~~Any physical device~~

why this, J for linear polarized light is

$$J_H = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, J_V = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

General elliptical polarized light $\rightarrow \begin{bmatrix} E_{0x} e^{i\delta_x} \\ E_{0y} e^{i\delta_y} \end{bmatrix}$

generic linear polarized light

$$\Rightarrow \begin{bmatrix} \cos \psi \\ \sin \psi \end{bmatrix}$$

Same ψ we saw here!
As for lin pol. only the
major axis would exist.

Any element that introduces a change of polarization
can be a 2×2 matrix.

$$\begin{bmatrix} E_{new x} \\ E_{new y} \end{bmatrix} = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \end{bmatrix}$$

Say if this was a half wave plate, it will change
the phase of element 1 & element 2 by 180°

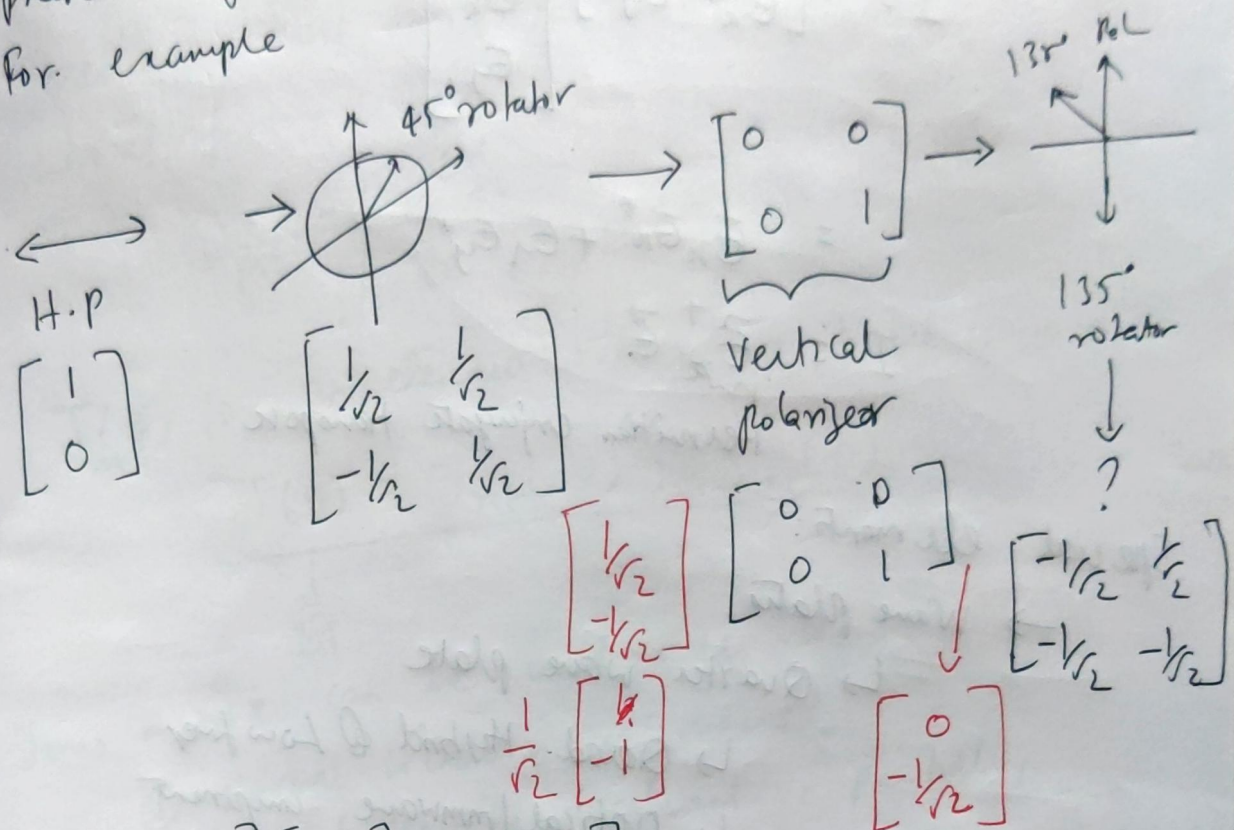
$$J_{HWP} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

For a generic polarizer which would just rotate
the angle by ψ , it would be

$$J_{rot} = \begin{bmatrix} \cos \psi & \sin \psi \\ -\sin \psi & \cos \psi \end{bmatrix}$$

Now, say we have a mirror, whose axis is
in the lab frame [lab x and lab y], but
it passes through a device that is oriented at
a different angle.

Some elements change the plane of polarization of the wave entirely, while other just add a phase change. The operations will be sequential.
For example



$$\begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \Rightarrow \text{light is extinct by half.}$$

Sequence

~~$$\begin{bmatrix} E_{outx} \\ E_{outy} \end{bmatrix} = \begin{bmatrix} \cos(45^\circ) & \sin(45^\circ) \\ -\sin(45^\circ) & \cos(45^\circ) \end{bmatrix} \begin{bmatrix} \cos(90^\circ) & \sin(90^\circ) \\ -\sin(90^\circ) & \cos(90^\circ) \end{bmatrix} \begin{bmatrix} E_x \\ E_y \end{bmatrix}$$~~

$$\begin{bmatrix} E_{outx} \\ E_{outy} \end{bmatrix} = \begin{bmatrix} \cos(135^\circ) & \sin(135^\circ) \\ -\sin(135^\circ) & \cos(135^\circ) \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(45^\circ) & \sin(45^\circ) \\ -\sin(45^\circ) & \cos(45^\circ) \end{bmatrix} \begin{bmatrix} E_x \\ E_y \end{bmatrix}$$

Intensity can be given as:

$$I = \begin{bmatrix} E_x^* & E_y^* \end{bmatrix} \begin{bmatrix} E_x \\ E_y \end{bmatrix}$$

$$= E_x E_x^* + E_y E_y^*$$

$$I = \vec{E}^\dagger \cdot \vec{E}$$

Hermitian conjugate transpose

Special elements:

→ Wave plates

↳ Quarter wave plate

↳ Quad. Hybrid @ Low freq

↳ Optical/mmwave component

at high freq

Introduces phase shift of $\phi = \pi/2$ between E_x and E_y components.

$$J_{WP} = \begin{bmatrix} 1 & 0 \\ 0 & e^{-j\pi/2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -j \end{bmatrix}$$

→ Half wave plate

introduces shift of $\lambda/2 = \pi$.

$$J_{HWP} = \begin{bmatrix} 1 & 0 \\ 0 & e^{-j\pi} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

A generic medium, that changes polarization of x and y components differently is called a birefringent medium. Most astrophysical plasmas are birefringent.

$$J_{\text{pol}} = \begin{bmatrix} e^{i\phi/2} & 0 \\ 0 & e^{-i\phi/2} \end{bmatrix}$$

$J(\theta)$ for an element rotated w.r.t lab frame is

$$J(\theta) = J_{\text{rot}}(-\theta) J(\phi) J_{\text{rot}}(\theta)$$

Jones matrix can be used only when

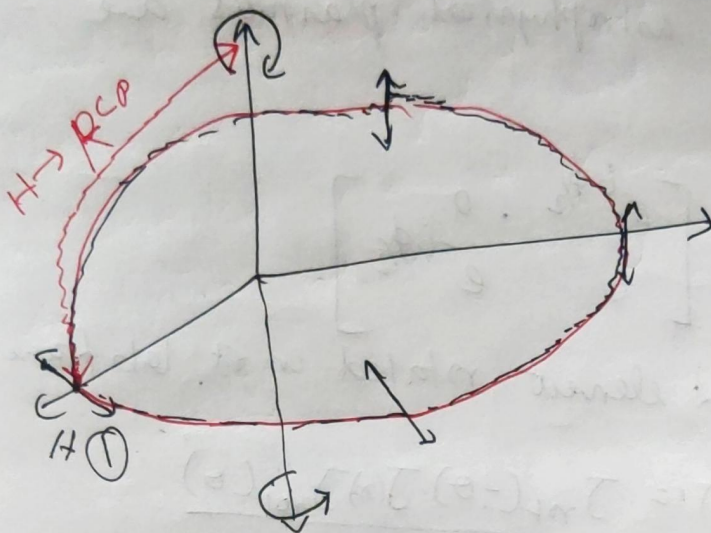
- ① Normal incidence is considered
- ② Can not describe incoherent or partially coherent light.

In other cases we will use Müller matrices. These matrices deal with how the Stokes parameters change when using a polarizing element.

$$\vec{M}_{\text{pol}} = \begin{bmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & m_{23} \\ m_{30} & m_{31} & m_{32} & m_{33} \end{bmatrix} \begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix} \Rightarrow \begin{bmatrix} S_0' \\ S_1' \\ S_2' \\ S_3' \end{bmatrix}$$

$$\begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix} \xrightarrow{\vec{S}} \begin{bmatrix} M_1 \\ M_2 \\ M_3 \end{bmatrix} \xrightarrow{\vec{S}'} \begin{bmatrix} S_0' \\ S_1' \\ S_2' \\ S_3' \end{bmatrix}$$

Transformations on the polycare sphere



$\longleftrightarrow \Rightarrow$ Rot (90°)
 $\longleftrightarrow \Rightarrow$ H.W.P $\Rightarrow$

$\longleftrightarrow \Rightarrow$ Q.W.P $\Rightarrow$
 $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & e^{-j\pi/2} \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & e^{-j\pi/2} \end{bmatrix}^* \Rightarrow ?$
 What is the path this will take?
Homework assignment

Paraxial angle

Source polar axis
 telescope beam

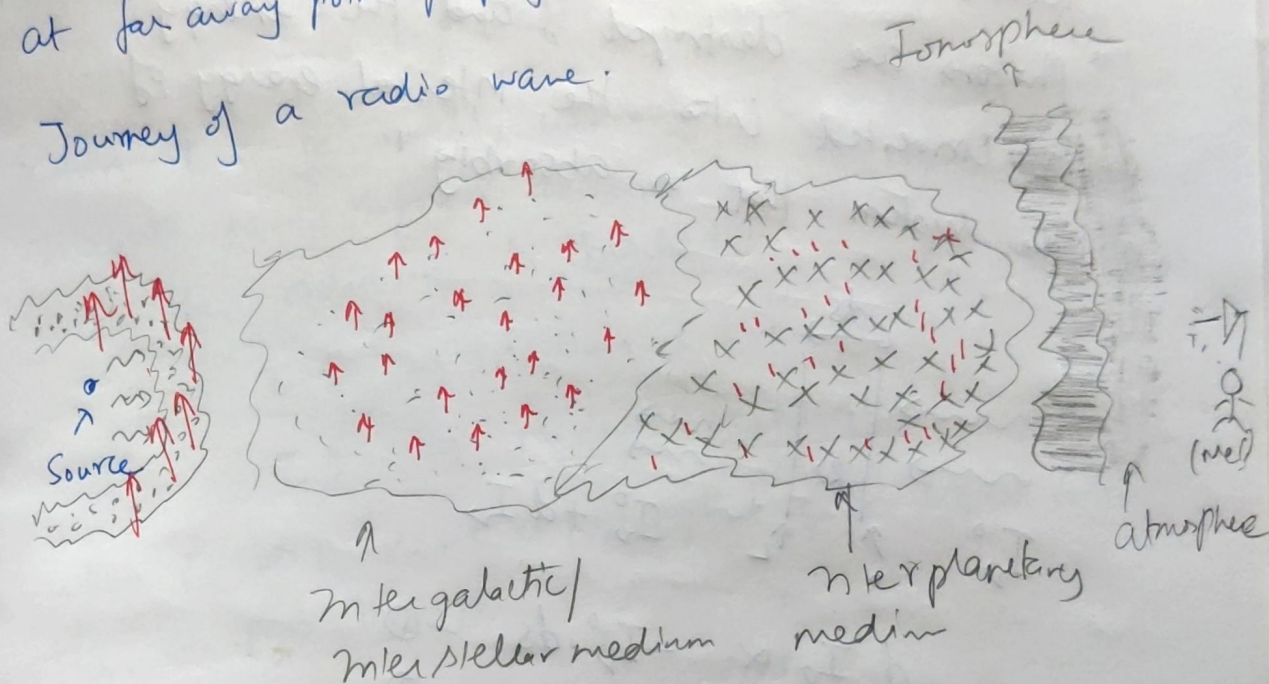
$$V_x = G(E_x \cos \psi_p + E_y \sin \psi_p)$$

$$V_y = G(E_x \sin \psi_p + E_y \cos \psi_p)$$

Propagation of Radiation

→ How does the radio waves that originate at far away point propagate.

Journey of a radio wave.



Radiation Transfer

How the waves travel through ~~space~~ a medium

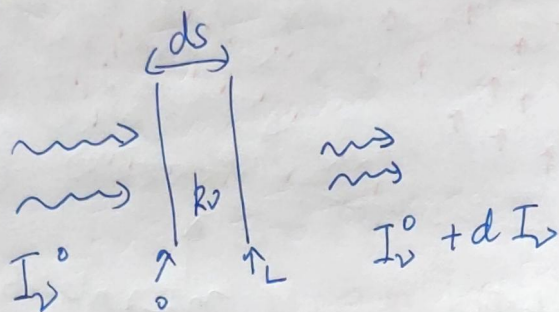
Radiation $\rightarrow$ Matter $\rightarrow$ Intensity

- $\hookrightarrow$ absorb [absorption lines, interstellar extinction]
- $\hookrightarrow$ emit [photon leaked away]
- $\hookrightarrow$ absorb & re-emit [fluorescence]
- $\hookrightarrow$ Scatter, change direction [blue sky]

all these interactions are microscopic but have an effect on the macroscopic propagation

① Absorption

↓
 leads to attenuation of light.
 " photons are destroyed, and the energy is converted into the internal energy of the system that is absorbing "



$$dI_v \propto -I_v^0 \cdot ds \quad \text{--- ①}$$

$$dI_v = -k_v I_v^0 ds \quad \text{--- ②}$$

k_v medium's ability to absorb at freq ν , per m.

k_v (cm^{-1}) → absorption coefficient

by rearranging ②, we will get

$$\frac{dI_v}{I_v} = -k_v ds$$

integration in both sides,

$$\ln \left[\frac{I_v}{I_v^0} \right]_{I_v^0}^{I_v} = \int_{L}^{0} -k_v ds$$

$$\ln \left(\frac{I_v}{I_v^0} \right) = - \int_0^L k_v ds$$

$\int_0^L k_v ds \rightarrow \text{optical depth.}$

$$I_v = I_0 \cdot \exp\left[-\int_0^L k_v ds\right]$$

$$I_v = I_0 \exp(-\tau_v)$$

when τ_v is small, almost all incident intensity passes through the medium. when it is large, then radiation is blocked.

$\Rightarrow \tau \downarrow \rightarrow \text{medium is transparent}$

$\tau \uparrow \rightarrow \text{medium is opaque}$

when $\tau_v \Rightarrow 1$, media is foggy, but some light from bly can pass through.

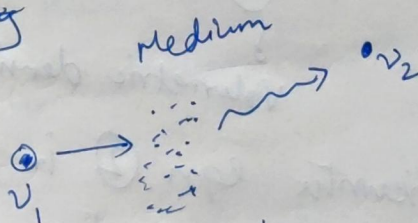
k_v, τ_v are frequency dependent.

$\Rightarrow \tau_v \sim 0$ in certain freq, while ∞ at other freq.

~~The cause of scattering~~

Scattering.

Photon



ν_1 & ν_2 are different frequencies.

This can also lead to decrease in intensity in the direction of the observer.

Both abs. and scattering at a macroscopic level depend on the material present in the medium & also the number of particles there.

Density of particles can be

① number density (n) m^{-3}

or

② mass density (ρ) kg m^{-3}

So, defining the abs. coefficient, we can describe it

as

$$K_v = \underbrace{\sigma_v n}_{\text{Cross section (cm}^2\text{)}} \cdot \underbrace{\rho}_{\text{particle density}} \quad \text{--- (3)}$$

$$\therefore dI_v = -I_v^0 \sigma_v n ds.$$

When we have a mass density associated, the absorbed radiation's amount can be defined in terms of "opacity" (K_v^m)

$$K_v^m = \frac{\sigma_v \cdot n}{\rho} \quad [\text{cm}^2 \text{g}^{-1}]$$

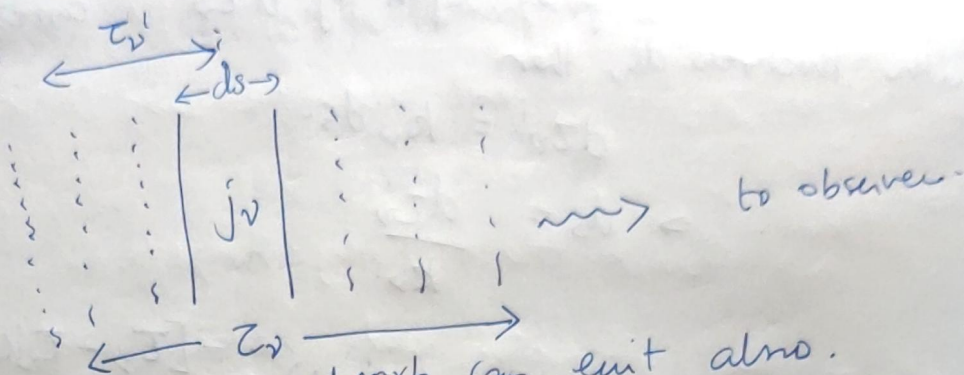
↓
Volumetric density.

If we rewrite eqn (2) in terms of K_v^m , we get

$$dI_v = -I_v^0 \sigma_v n \cdot ds$$

$$dI_v = -I_v^0 K_v^m \cdot \rho \cdot ds$$

Emission



a medium that can absorb. can emit also.
Emission by an infinitesimally thin layer can be defined using:

$$dI_v' = \underbrace{j_v}_{\text{emissivity}} \cdot \underbrace{ds}_{\text{layer thickness}}$$

↓
Emission added by ds
 $J S^{-1} Hz^{-1} sr^{-1} m^{-3} \rightarrow \text{unit of emissivity.}$

⊗ We consider the emission only from the medium.
Absorption has to be considered still as that emitted radiation has to travel through the medium to get out. Now, there is an optical depth of the far side, Z_v' . Emission from far side must travel thru' this layer too.

$$\Rightarrow dI_v' = j_v \cdot ds \cdot e^{-(Z_v - Z_v')}$$

$Z_v \rightarrow$ total optical depth.

$\therefore$ Total emission from the medium is

$$I_v = \sum_i dI_v'(i) = \sum_i j_v(i) ds \cdot \exp[-(Z_v - Z_v')]$$

if an incremental optical depth of dz is associated with thickness ds , then

$$dz' = k_v ds$$

$$\Rightarrow ds = \frac{dz'}{k_v}$$

$$\therefore I_v = \sum \frac{j_v(i)}{k_v(i)} d\tau_v(i) e^{-[\tau_v - \tau_v'(0)]}$$

replacing the sum with an integral

$$I_v = \int_0^{\tau_v} \frac{j_v}{k_v} dz' e^{-(\tau_v - \tau_v')}$$

$$= \int_0^{\tau_v} \frac{j_v \cdot dz' \cdot e^{-\tau_v} - j_v dz' e^{+\tau_v'}}{k_v}$$

if j_v, k_v are position independent

This is valid as position info. of this change is not available, and we just use the avg. values.

$$= \left[\frac{j_v}{k_v} \right] e^{-\tau_v} \int_0^{\tau_v} e^{\tau_v'} d\tau_v'$$

$$= \frac{j_v}{k_v} e^{-\tau_v} [e^{\tau_v'}]_0^{\tau_v}$$

$$= \left[\frac{j_v}{k_v} \right] \cdot e^{-\tau_v} [e^{\tau_v} - e^0]$$

$$I_v = \frac{j_v}{k_v} [1 - e^{-\tau_v}]$$

$$S = \frac{j\nu}{k\nu} \rightarrow \text{Source function}$$

$$\Rightarrow I_\nu = \frac{j\nu}{k\nu} [1 - e^{-\tau_\nu}]$$

when $\tau_\nu \rightarrow \infty$, $e^{-\tau_\nu} \rightarrow 0$.

$\Rightarrow$ ~~$I_\nu = \frac{j\nu}{k\nu}$~~ when optical depth is very small, then, $\tau_\nu \downarrow$

$\Rightarrow$ Then $e^{-\tau_\nu}$ can be expanded into the series,

$$e^{-\tau_\nu} \approx 1 - \tau_\nu$$

$$\begin{aligned} \Rightarrow I_\nu &= S_\nu [1 - e^{-\tau_\nu}] \\ &= S_\nu [1 - 1 + \tau_\nu] \end{aligned}$$

$$\boxed{I_\nu = S_\nu \cdot \tau_\nu} \text{ optically thin limit.}$$

If the absorption was constant over the medium with thickness of L , then $\tau_\nu = k_\nu L$

$$\Rightarrow I_\nu = S_\nu \cdot k_\nu \cdot L = \frac{j\nu}{k_\nu} \cdot k_\nu \cdot L \approx j\nu \cdot L$$

$\Rightarrow$ Intensity depends only on the emission.

$\Rightarrow$ emission intensity is $\propto$ thickness of the medium.

If τ was ∞ , then

$$I_\nu \approx S_\nu$$

$\Rightarrow$ Absorption & emission compete, and intensity dominated by what dominates the other.

For optically thick regime, the source fn. will define the intensity. [In thermal radiation case, it is equal to $B(T)$]
 $\rightarrow$ Planck black body brightness.

Based on the absorption and emission coeff, we can write a general radiative transfer equation

$$I_\nu = I_\nu^0 e^{-\tau_\nu} + \int_0^{\tau_\nu} \frac{j_\nu}{k_\nu} d\tau'_\nu \exp[-(\tau_\nu - \tau'_\nu)]$$

Eval. this integral as before

$$I_\nu = I_\nu^0 e^{-\tau_\nu} + S_\nu (1 - e^{-\tau_\nu})$$

if ~~source~~

for $\tau = 0$, then

$I_\nu = I_\nu^0 \rightarrow$ medium is completely transparent

for $\tau = \infty$, then

I_ν^0 cannot pass and all we have is only $S_\nu \rightarrow$ emissivity of the medium
 absorption

if $I_\nu > S_\nu$, then ~~intensity I_ν increases as~~

~~we~~ I_ν decreases and emergent intensity is lower than incident

$S_\nu > I_\nu$, then emergent is higher than incident.

$\tau=1$ is where stellar photospheres are defined.

$I_0 \rightarrow S_0 \rightarrow B_0(\tau)$ which is why stars appear as black body emitters.

for a case of $\tau=1$,

$$I_\nu = I_0 \cdot e^{-\tau_\nu} + S_0(1 - e^{-\tau_\nu})$$

$$\begin{aligned} \tau_\nu &= 1 \\ \Rightarrow I_\nu &= I_0 e^{-1} + S_0(1 - e^{-1}) \quad (e^{-1} \approx 0.37) \\ &= (0.37)I_0 + S_0(0.63) \end{aligned}$$

if $I_0 = S_0$, then

$$I_\nu = 0.37S_0 + 0.63S_0$$

$\Rightarrow I_\nu = S_0 \rightarrow$ all photons absorbed
are replaced

Example: LTE atmosphere at same temp as b.g. radiation.

when $I_0 > S_0$, then, output intensity is more closer to S_0 , $\Rightarrow I_0$ is absorbed & $I_{\text{emitted}} < I_0$

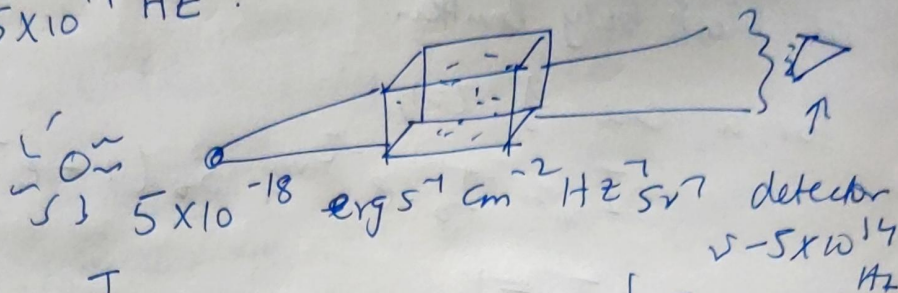
Examples: absorption lines in stellar spectra, molecular cloud in front of a hot star
medium is an absorber

but when $I_0 < S_0$, then there is no background radiation seen. Medium appears as an emitter

Example problem

Beam Intensity = $I_\nu = 5 \times 10^{-18} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1}$

freq = $5 \times 10^{14} \text{ Hz}$



when spotlight is turned off, detector still measures $2 \times 10^{-8} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1}$

$$\left. \begin{array}{l} I_{\text{det}} \\ \sim 3 \times 10^{-8} \\ \text{erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1} \end{array} \right\}$$

① what is τ_ν & S_ν of the gas when using the pure emitter case,

$$I_\nu \approx S_\nu (1 - e^{-\tau_\nu})$$

$$2 \times 10^{-8} \approx S_\nu (1 - e^{-\tau_\nu})$$

when $I_{\nu \text{ on}}$, then

$$I_{\nu \text{ on}} = I_{\nu \text{ on}} e^{-\tau} + S_\nu (1 - e^{-\tau})$$

$$I_{\nu \text{ on}} = I_{\nu \text{ off}} e^{-\tau} + I_{\nu \text{ off}}$$

$$2 \times 10^{-8} = 5 \times 10^{-18} e^{-\tau} + 5 \times 10^{-18}$$

$$I_{\nu \text{ on}} - I_{\nu \text{ off}} = I_{\nu \text{ off}} e^{-\tau}$$

$$e^{-\tau} = \frac{I_{\nu \text{ on}} - I_{\nu \text{ off}}}{I_{\nu \text{ off}}} = \frac{1 \times 10^{-8}}{5 \times 10^{-18}}$$

$$e^{-\tau} = 0.2 \times 10^{-8} \times 10^{18}$$

$$e^{-\tau} = 0.2 \times 10^{10} \Rightarrow \tau = -\ln(0.2 \times 10^{10}) = 1.61$$

6) Cross section for interaction b/w photons & gas particles $\sim 5.7 \times 10^{-18} \text{ cm}^2$. Cube is 50 cm in length & has uniform density. What is the number of gas particles?

$$\tau = k_v \cdot L$$

$$= n \cdot \sigma_v \cdot L$$

$$\tau = n \cdot \frac{5.7 \times 10^{-18} \text{ cm}^2}{(\text{cm}^2)} \cdot 50 \text{ cm}$$

$$\tau = 1.61 \Rightarrow n = \frac{1.61}{5.7 \times 10^{-18} \cdot 50 \text{ cm}}$$

$$n = 5.6 \times 10^{15} \text{ gas particles.}$$

Source function $\rightarrow 2 \times 10^{-8} = S_v (1 - e^{-1.61})$

$$2 \times 10^{-8} = S_v$$

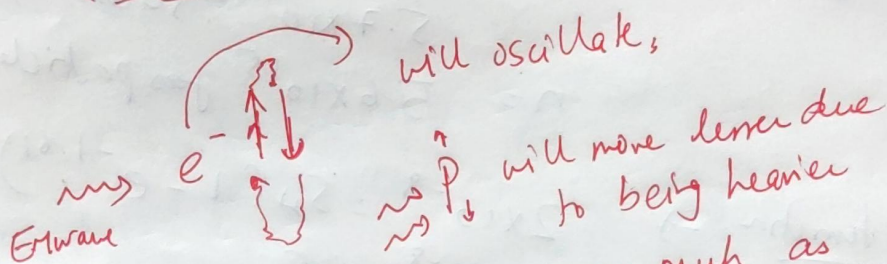
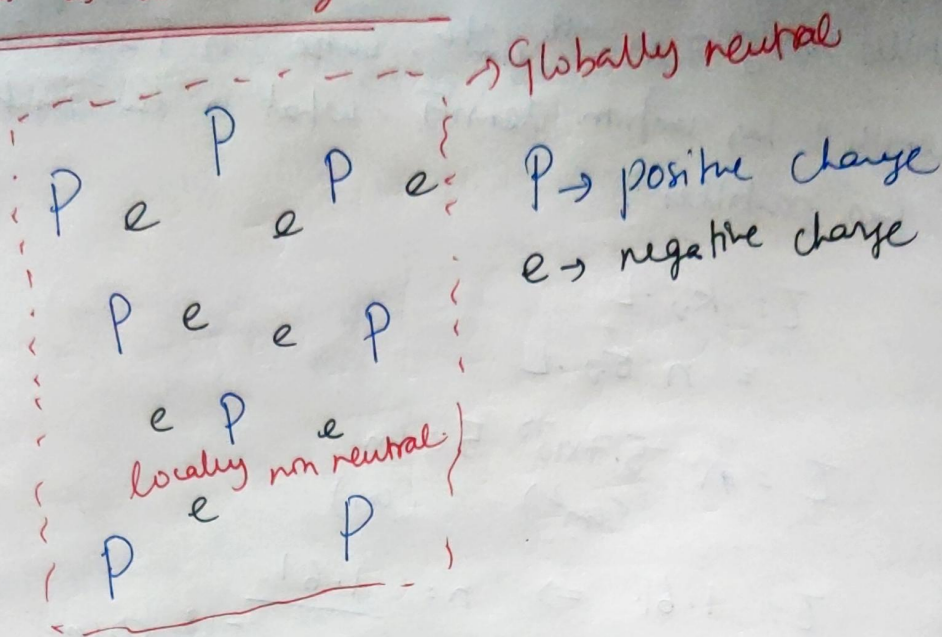
$$\frac{2 \times 10^{-8}}{1 - e^{-1.61}}$$

$$\Rightarrow S_v = 2.5 \times 10^{-8} \text{ erg cm}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1}$$

7) Source fm is smaller than intensity of the object.

$\Rightarrow$ gas is a weaker source. $\Rightarrow$ ~~source~~ gas is colder than the object's radiation

Propagation in an Ionized Medium



Protons don't affect EM waves as much as electrons.

~~From~~ electrons → ISM, IGM, Heliosphere etc.

Cause phenomenon →

- ① Plasma frequency
- ② Dispersion
- ③ Faraday rotation

① Plasma frequency:

$$\left. \begin{aligned} \nabla \times B &= \mu_0 J + \mu_0 \epsilon_0 \frac{\partial E}{\partial t} \\ \nabla \times E &= - \frac{\partial B}{\partial t} \end{aligned} \right| \begin{aligned} m_e \cdot \frac{dv}{dt} &= -eE \\ J &= \sigma E \end{aligned}$$

for a plane wave ~~$E = E_0 e^{i(kx - \omega t)}$~~ $E = E_0 e^{i(kz - \omega t)}$
 $E_z = E_0 e^{i(kz - \omega t)}$

in 1D, $E = E_0 \exp(kz - \omega t)$

$\frac{d}{dt} \rightarrow -i\omega$ — use this in motion equation

$$m_e \frac{dv_e}{dt} = -eE$$

$$-i\omega m_e v_e = -eE$$

$$v_e = \frac{1}{i\omega m_e} eE$$

current density $\rightarrow J = \underbrace{-ne}_\text{amount of total charge} \underbrace{eV}_\text{travelling at velocity}$

$$\Rightarrow J = -n_e \cdot e \cdot \frac{1}{i\omega m_e} eE$$

$$= \frac{-n_e \cdot e^2}{\omega m_e} \cdot \frac{1}{i} \cdot E$$

$$J = i \left(\frac{n_e e^2}{\omega m_e} \right) \cdot E \quad \text{— using this in ~~new~~ M/ω eqn.}$$

Take use of Faraday's law

$$\nabla \times (\nabla \times E) = -\frac{\partial}{\partial t} (\nabla \times B)$$

$$\nabla \times (\nabla \times E) = -\frac{\partial}{\partial t} \left[\mu_0 J + \mu_0 \epsilon_0 \cdot \frac{\partial E}{\partial t} \right]$$

$$\nabla(\nabla \cdot E) - \nabla^2 E = -\left(\mu_0 \frac{\partial J}{\partial t} + \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2}\right)$$

$$\nabla(\nabla \cdot E) - \nabla^2 E = -\left(\mu_0 \cdot \frac{\partial}{\partial t} \left(\frac{j n e e^2}{\omega m_e}\right) E + \mu_0 \epsilon_0 \cdot \frac{\partial^2 E}{\partial t^2} e^{j(\omega t + k z)}\right)$$

$\rightarrow \frac{\partial}{\partial x} \rightarrow jk$

$$\nabla \left[\nabla \cdot E e^{j(\omega t + k z)} \right] - \nabla^2 E e^{j(\omega t + k z)}$$

$$\cancel{\nabla(\nabla \cdot E)} \quad k^2 (E) - k^2 E = - \left[\left(\mu_0 \cdot \frac{j n e e^2}{\omega m_e} \cdot j \omega \right) E + \mu_0 \epsilon_0 \cdot (-1) \omega^2 E \right]$$

$$k^2 E - \underbrace{k^2}_{\downarrow 0} (E) = \mu_0 \left(\frac{-n e e^2 \cdot \omega E}{\omega m_e} - \omega^2 \mu_0 \epsilon_0 E \right)$$

$$\Rightarrow k^2 E = \frac{e^2 n e \cdot \mu_0 \cdot E}{m_e} - \omega^2 \mu_0 \epsilon_0 E$$

$$k^2 = \left[\omega^2 \mu_0 \epsilon_0 + \frac{e^2 n e \mu_0}{m_e} \right]$$

$$k^2 = + \left[\frac{\omega^2}{c^2} + \frac{e^2 n e \mu_0}{m_e} \right]$$

$$k^2 c^2 = + \left[\omega^2 + \frac{e^2 n e}{m_e \cdot \epsilon_0} \right] \quad \begin{matrix} \text{plasma} \\ \text{freq} \end{matrix}$$

$$k^2 c^2 + \omega_p^2 = \omega^2 \quad \text{--- Dispersion relation in plasma}$$

$$v_p = \frac{\omega}{k}$$

$$= \frac{\omega}{\frac{\omega}{c \sqrt{1 - \left(\frac{\omega_p^2}{\omega^2}\right)}}} = \frac{c}{\sqrt{1 - \left(\frac{\omega_p}{\omega}\right)^2}}$$

$$v_g = \frac{d\omega}{dk}$$

diff dispersion relation w.r.t k ,

$$2\omega \frac{d\omega}{dk} = 2kc^2$$

$$\omega \frac{d\omega}{dk} = \frac{\omega}{c} \sqrt{1 - \left(\frac{\omega_p}{\omega}\right)^2} \cdot c^2$$

$$\Rightarrow \frac{d\omega}{dk} = c \sqrt{1 - \left(\frac{\omega_p}{\omega}\right)^2}$$

group velocity

Now, we see that the group velocity is frequency dependent. How would this delay the EM waves passing through the plasma?

Assume the wave is transmitting a distance of ds , the time taken can then be written as,

$$dt = \frac{ds}{v_g} = \frac{ds}{c \sqrt{1 - \left(\frac{\omega_p}{\omega}\right)^2}}$$

$$= \frac{1}{c} \left[1 - \left(\frac{\omega_p}{\omega}\right)^2 \right]^{-1/2} \cdot ds$$

$$\left[1 - \left(\frac{\omega_p}{\omega}\right)^2\right]^{-1}$$

→ this can be expanded out to the first term as

$$\approx \left[1 + \frac{1}{2} \left(\frac{\omega_p}{\omega}\right)^2\right] + \dots$$

Substituting this in group delay,

$$dt = \frac{ds}{c} \left[1 + \frac{1}{2} \left(\frac{\omega_p}{\omega}\right)^2\right]$$

$$\boxed{\frac{ds}{c}} + \underbrace{\frac{1}{2} \left(\frac{\omega_p}{\omega}\right)^2}_{\Delta t}$$

↓ vacuum / free space delay

$$\Rightarrow d(\Delta t) = \frac{1}{2} \cdot \left(\frac{\omega_p}{\omega}\right)^2 \cdot \frac{ds}{c}$$

Subs. ω_p in this eqn

$$\omega_p^2 = \frac{n_e e^2}{\epsilon_0 \cdot m_e}$$

$$= \frac{1}{2} \frac{n_e e^2}{\epsilon_0 m_e} \cdot \frac{1}{\omega^2} \cdot \frac{ds}{c}$$

But n_e can vary with distance so we write n_e as a function of distance implicitly.

$$\Rightarrow d(\Delta t) = \frac{1}{2} \frac{n_e(s) e^2}{\epsilon_0 m_e} \cdot \frac{1}{\omega^2} \cdot \frac{ds}{c}$$

Re arranging,

$$d(\Delta t) = \frac{1}{2} \cdot \frac{e^2}{\epsilon_0 m_e} \cdot \frac{1}{\omega^2 c} (n_e \cdot ds)$$

Integrating from source to observer,

$$\Delta t = \frac{1}{2} \frac{e^2}{\epsilon_0 m_e c} \cdot \frac{1}{\omega^2} \underbrace{\int_0^D n_e(s) ds}_{\text{Dispersion Measure}}$$

$$\Delta t = \frac{1}{2} \left(\frac{e^2}{\epsilon_0 m_e c} \right) \cdot \frac{DM}{\omega^2}$$

$$\Rightarrow \left\{ \begin{array}{l} \omega = 2\pi f \\ \Delta t = \frac{f^2}{4\pi \epsilon_0 m_e c} \cdot \frac{DM}{f^2} \end{array} \right\} \begin{array}{l} \text{more dispersion} \\ \text{at long wave lengths.} \end{array}$$

How does the plane of polarization rotate?

Thermal & Non-Thermal emission

Raleigh Jeans

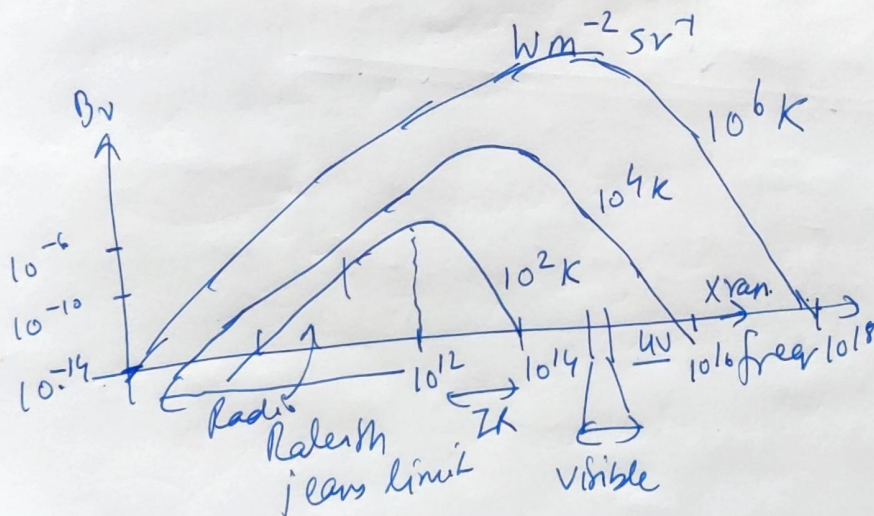
$$B_\nu(T) \approx \frac{2kT}{\lambda^2}, \text{ when } \frac{h\nu}{kT} \ll 1$$

or

$$B_\nu(T) \approx \frac{2k\nu^2}{c^2} \cdot T \quad \left. \vphantom{\frac{2k\nu^2}{c^2} \cdot T} \right\} \text{ tail of the Planck's law}$$

Flux $\rightarrow \int_{\text{solid angle}} B_\nu(T) \cdot d\Omega$
(Jy)

$$= B_\nu(T) \Omega = \left\{ \frac{2kT}{\lambda^2} \right\} \underbrace{\Omega}_{\text{steradian}}$$



$$F = \frac{2kT_B}{\lambda^2} \Omega, \text{ if } T \rightarrow T_B \text{ brightness temperature}$$

$$\frac{F \cdot \lambda^2}{2k\Omega} = \underline{T_B} [K]$$

Question

at $f = 100 \text{ MHz}$

Cygnus A has a typical flux density of $10,000 \text{ Jy}$. The size of the source is $2' \times 4'$. What is the brightness temperature?

$$1' = \left(\frac{1}{60}\right)^\circ, \quad 150^\circ - 40,000 \text{ sq. deg.}$$

150 degrees

$$150 \rightarrow 3282.8 \text{ sq. deg.}$$

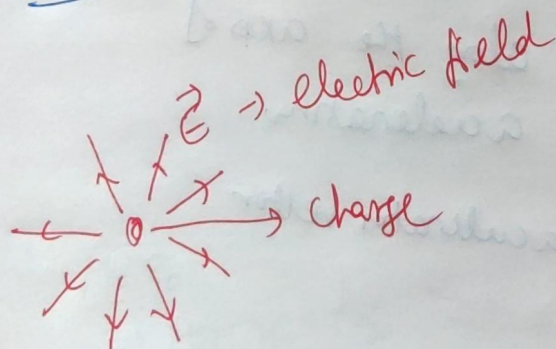
$$1 \text{ sq. deg} \rightarrow \left(\frac{180}{\pi}\right)^2 \text{ deg}^2.$$

at 100 MHz Cyg A has $S_\nu \approx 2000 \text{ Jy}$.

Does this have the blackbody spectrum?

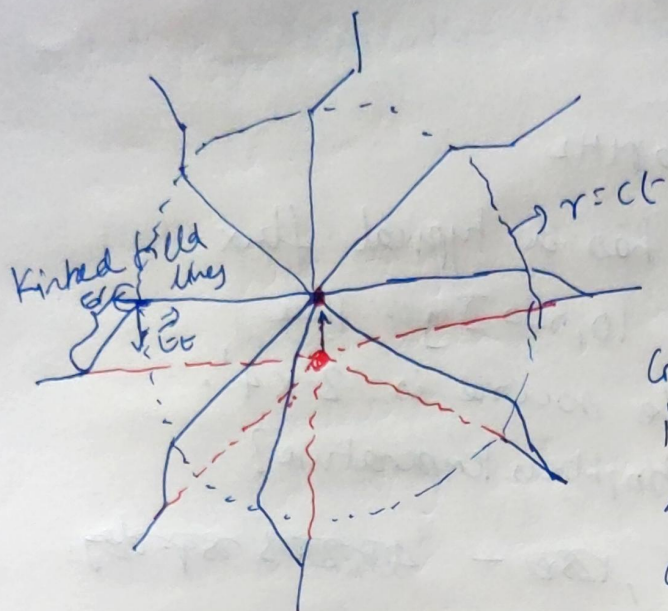
Electromagnetic radiation

Radiation from accelerated pt. charge



$\vec{E}$ is varying radially.

If this charge is slightly given an acceleration, the field lines would change as shown:

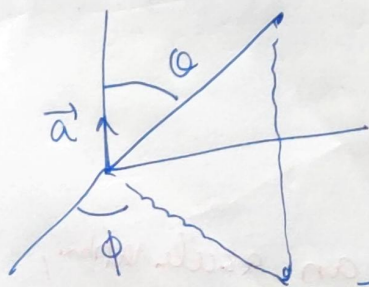


for the change in the
implication after the
initial acceleration
shows up at any point
at the speed of light.

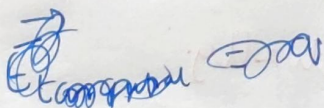
But an observer who is far away will still see
the original electric field of the particle. The
field in the transition region, or in the "kink"
region has both radial and tangential
components. The Tangential component is associated
with the ~~the~~ EM wave.

$E_t \rightarrow$ varies with direction

$E_t \rightarrow 0$, along the axis of
acceleration



$\vec{a}$ acceleration vector



$$\vec{E}_t \propto \frac{a \sin \theta}{r}$$

$$\vec{E}_t = \frac{q \cdot a \cdot \sin \theta}{r \cdot c^2} \cdot \frac{1}{4\pi \epsilon_0 c^2}$$

$$E_{\text{rad}} = \frac{q \cdot a \sin \theta}{4\pi\epsilon_0 c^2 R} \quad E_t \rightarrow \text{radiated field}$$

Power from this can be written using the Poynting

flux.
 $\Rightarrow$ But $S = \frac{1}{\mu} (\vec{E} \times \vec{B})$

But $|B_{\text{rad}}| = \frac{|\vec{E}_{\text{rad}}|}{c}$

$\Rightarrow$ But $\vec{E} \perp \vec{B}$ as $\vec{r}$ is the zone
 when radiation is happening $\Rightarrow E \perp B \perp \hat{n}$
 direction of wave prop.

$$\Rightarrow S = \frac{1}{\mu_0} \cdot \frac{E^2}{c} = \frac{1}{\mu_0 c} \cdot \frac{q^2 a^2 \sin^2 \theta}{16\pi^2 \epsilon_0 c^2 R^2}$$

$$= \frac{q^2 a^2 \sin^2 \theta}{16\pi^2 \cdot \mu_0 \epsilon_0 c^3 R^2}$$

But $\frac{1}{\mu_0 \epsilon_0} = c^2$

$\Rightarrow \mu_0 = \frac{1}{c^2 \epsilon_0}$

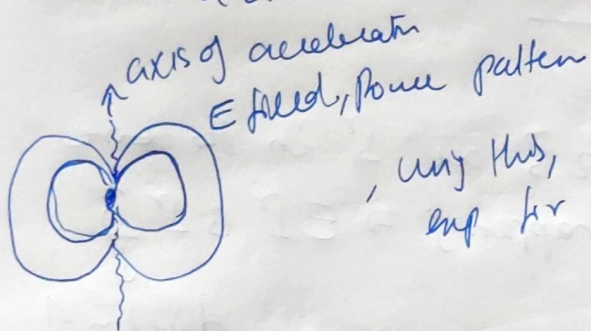
$\mu_0 = \frac{1}{\epsilon_0 c^2} \Rightarrow$

$$\frac{q^2 a^2 \sin^2 \theta}{16\pi^2 \cdot \frac{1}{\epsilon_0 c^2} \cdot c^3 R^2}$$

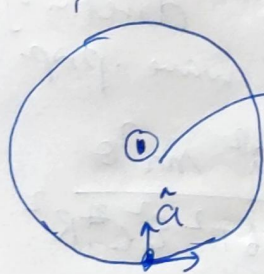
$$= \frac{q^2 a^2 \sin^2 \theta}{16\pi^2 \epsilon_0 c^3 R^2} = \frac{q^2 a^2 \sin^2 \theta}{16\pi^2 \epsilon_0 c^3 R^2}$$

$\therefore$ Poynting flux, $S(r)$ from a radiated particle will:

- ① have only latitude component.
- ② is ~~independently~~ the direction of acceleration
- ③ at any radius, it is constant at azimuth



using this, we can write an exp for power.



$$P = \int_{\text{sphere}} |\vec{S}| \cdot dA$$

$$P = \frac{q^2 a^2}{16\pi^2 c^3 \epsilon_0} \int_0^{2\pi} \int_0^\pi \frac{\sin^2 \theta}{r^2} \cdot r \sin \theta \, d\theta \cdot r \, d\phi$$

$$= \frac{q^2 a^2}{16\pi^2 c^3 \epsilon_0} \cdot \int_0^{2\pi} \int_0^\pi \frac{\sin^2 \theta}{r^2} \cdot r \sin \theta \, d\theta \cdot r \, d\phi$$

$$= \frac{q^2 a^2}{16\pi^2 c^3 \epsilon_0} \int_0^{2\pi} \int_0^\pi \sin^3 \theta \, d\theta \cdot d\phi$$

$$= \frac{q^2 a^2}{16\pi^2 c^3 \epsilon_0} \cdot \int_0^{2\pi} d\phi \int_0^\pi \sin^3 \theta \, d\theta$$

$$\frac{q^2 a^2}{16\pi^2 c^3 \epsilon_0} [2\pi] \int_0^\pi \sin^3 \theta \, d\theta$$

$$\Rightarrow P = \frac{q^2 a^2}{16 \pi^2 c^3 \epsilon_0} \cdot [2\pi] \cdot \left[\frac{8}{3}\right]$$

$$= \frac{2 \cancel{[q^2 a^2]} \cancel{[8\pi c^3 \epsilon_0]}}{3 \cancel{[16\pi^2 c^3 \epsilon_0]}} \cdot \boxed{\frac{2 \cancel{[q^2 a^2]} \cancel{[8\pi c^3 \epsilon_0]}}{3 \cancel{[16\pi^2 c^3 \epsilon_0]}}}$$

$$= \frac{q^2 a^2}{24 \pi^2 c^3 \epsilon_0} \left[\frac{8\pi}{3}\right] \quad \text{Larmor's Formula}$$

$$P \propto a^2 \quad = \frac{q^2 a^2}{6 \pi c^3 \epsilon_0}$$

Acceleration occurs due to

① Coulomb interaction

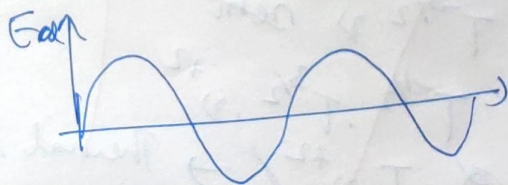
~~② Coulomb interaction~~

② $e^- - p^+$ produce EM radiation

$\downarrow$
is less massive $\Rightarrow$ gets more accelerated.

As interaction is Coulombic, we can find out what will be the acceleration with time. This will give us $E_{\text{tan}} / E_{\text{rad}} \Rightarrow FT \Rightarrow E_{\text{tan}} / E_{\text{rad}}(\nu)$

If $\vec{a}$ was true harmonic then, E_{rad} will also vary with time at same freq.



$$\Rightarrow E_{\nu} \quad \left| \begin{array}{c} \text{---} \uparrow E_0 \\ \omega \end{array} \right.$$

thermal bremsstrahlung.

→ electrons accelerated in plasma due to thermal motion

→ interacts coulombically with other electrons/ ions

→ e^- - ion radiation is wideband at all freqs.

⇒ emissivity j_ν

$$j_\nu^{\text{th}} = \frac{8}{3} \left(\frac{\pi}{2} \right)^{1/2} \cdot \frac{Z^2 e^6}{m_e^{3/2} c^3 (4\pi\epsilon_0)^3} \cdot n_e n_i T^{-1/2} e^{-h\nu/kT}$$

at $h\nu \ll kT$,

$$j_\nu^{\text{th}} \propto n_e n_i T^{-1/2}$$

$$\begin{aligned} -\frac{1}{2} - 1 \\ -\frac{1-2}{2} \\ = -\frac{3}{2} \end{aligned}$$

but we know that $\frac{j_\nu}{\alpha_\nu} = \frac{B(\nu)}{\downarrow}$
Rayleigh Jeans for $h\nu \ll kT$

$$\& B(\nu) = \frac{2kT\nu^2}{c^2}$$

$$\Rightarrow \alpha_\nu^{\text{th}} \propto n_e n_i T^{-3/2} \cdot \nu^{-2}$$

Solution

$$S_{100\text{MHz}} = 10,000 \text{ Jy}$$

$$S_{1000\text{MHz}} = 2000 \text{ Jy}$$

Source size $\rightarrow 2' \times 4'$

↓

Convert this first to steradians

$$1' \rightarrow \left(\frac{1}{60}\right)^\circ$$

$$\Rightarrow 2' \rightarrow \left(\frac{2}{60}\right)^\circ, \quad 4' \rightarrow \left(\frac{4}{60}\right)^\circ$$

$$\Rightarrow \text{Source size} \sim \left(\frac{2}{60}\right)^\circ \times \left(\frac{4}{60}\right)^\circ$$

~~Source~~ If we approx. the source region to be an ellipse, then, the area of the source in the sky plane will have a solid angle of

$$\theta_{\text{solid}} = \pi ab$$

$$= \pi \cdot \left(\frac{2}{60}\right)^\circ \cdot \left(\frac{4}{60}\right)^\circ = 7 \times 10^{-3} \text{ Sq. deg.}$$

In 1 Steradian, there is ~ 3283 sq. deg.

$$\Rightarrow \begin{array}{l} \text{1 sr} \rightarrow 3283 \text{ sq. deg} \\ \text{flux } \times \text{sr} \rightarrow 7 \times 10^{-3} \text{ sq. deg} \end{array}$$

$$\Rightarrow \Theta_{\text{sr}} = \frac{7 \times 10^{-3}}{3283} = 2.13 \times 10^{-6} \text{ sr}$$

Using Rayleigh Jeans, we know,

$$B_{\nu}(T) = \frac{2KT\nu^2}{c^2}$$

But flux density $S_{\nu} = \int B_{\nu}(T) d\Omega$

$$S_{\nu} = 1$$

$$\therefore S_{\nu} = \frac{2KT\nu^2}{c^2} \cdot \Omega$$

$$\Rightarrow T = \frac{S_{\nu} \cdot c^2}{2K\nu^2} \cdot \frac{1}{\Omega}$$

$$T_{100\text{MHz}} = \frac{10,000 \times (3 \times 10^8)^2}{2 \times 1.38 \times 10^{-23} \times (100 \times 10^6)^2} \cdot \frac{1}{(2.13 \times 10^{-6} \text{ sr})}$$

$$= \frac{1 \times 10^4 \times (3 \times 10^8)^2}{2 \times 1.38 \times 10^{-23} \times 10^{16}} \cdot \frac{1}{2.13 \times 10^{-6}} = 3 \times 10^{12} \text{ K}$$

$$\frac{9 \times 10^{16} \times 10^4}{2 \times 1.38 \times 10^{-23} \times 10^{16} \times 2.13 \times 10^{-6}}$$

$$= \frac{9 \times 10^{20}}{5.88 \times 10^{-29} \times 10^{16}} = \frac{9 \times 10^{20}}{5.88 \times 10^{-13}}$$

$$\frac{9 \times 10^4}{5.8 \times 10^{-29}} = \frac{1.53 \times 10^4 \times 10^{-26}}{5.8 \times 10^{-29}}$$

$$= 1.53 \times 10^{-22} \times 10^{29}$$

$$T_b = 1.53 \times 10^7$$

Similarly at 19 Hz,

$$T_b = 30,000 \text{ K.}$$

$\Rightarrow T_b$ is not constant with freq.

$\Rightarrow$ The source does not follow Rayleigh-Jeans law

